

Joohyun Lee

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RESEARCH INTERESTS

#RobotLearning #HumanRobotInteraction #AutonomousDriving #3DPerception #VisionLanguageAction
#EmbodiedAI #MultimodalLearning

EDUCATION

Georgia Institute of Technology, Atlanta, United States
Online Master of Science in Computer Science

Aug 2024 — Present
Overall GPA: 4.00/4.00

- Master's project: EvoReg: Versatile and Robust Point Cloud Registration via Multi-Stage Alignment

Sungkyunkwan University, Suwon-si, South Korea
Bachelor of Mechanical Engineering

Mar 2018 — Feb 2023
Cumulative GPA: 3.85/4.50

WORK EXPERIENCE

Hyundai Motor Company, Seoul, South Korea
Autonomous Driving Division
Research Engineer

Jan 2023 — Present

- **Hyundai RoboTaxi — LiDAR Processing:** Built a CUDA-accelerated LiDAR interface in Python, enabling high-throughput sensor data processing for RoboTaxi perception development and validation.
- **Atria AI — End-to-End Perception:** Designed and developed both camera-based and 3D end-to-end multi-object tracking models in Python, built a custom evaluation pipeline, and improved model performance through architectural redesign and refinement, contributing to the development of Hyundai Motor Group's proprietary autonomous driving model, Atria AI.

Samsung Research, Seoul, South Korea
Robot Center GEMS LAB
University Student Intern

Jul 2022 — Aug 2022

- Developed an action-recognition algorithm using IMU data collected from a wearable robot.
- Improved classification accuracy for boxing-movement recognition.

RESEARCH EXPERIENCE

EvoReg: Versatile and Robust Point Cloud Registration via Multi-Stage Alignment
Human-Augmented Analytics Group (HAAG), Georgia Institute of Technology
Advisors: Dr. Supratik Mukhopadhyay, Dr. Nick Lytle

Aug 2025 — Present

Developed a coarse-to-fine framework for rigid and non-rigid point cloud registration, with a focus on robustness to noise, partial overlap, and deformation. Contributed throughout the project, including model architecture design, baseline evaluation pipeline development, inference optimization, and manuscript preparation. Demonstrated strong performance across multiple registration benchmarks under a unified evaluation protocol. Submitted to ICLR 2027 (under review).

LatticeSpike: Real-Time Token-Level Hallucination Detection via Box Embeddings and Spiking Dynamics

Aug 2025 — Present

Human-Augmented Analytics Group (HAAG), Georgia Institute of Technology
Advisors: Dr. Supratik Mukhopadhyay, Dr. Nick Lytle

Contributed to a real-time hallucination detector that runs alongside a frozen LLM and scores every token as it is generated. Focused on the signal-extraction stage of the pipeline: implemented the causal per-step signal readout from the language model (hidden states, top-k next-token candidates, per-layer logit-lens KL divergence, attention entropy, and effective rank of layer representations). Designed the pipeline visualizations illustrating the signal-readout, box-embedding, and spiking-dynamics stages of the architecture. The resulting detector demonstrated strong performance across multiple hallucination-detection benchmarks. Submitted to ICLR 2027 (under review).

Indoor Autonomous Delivery Robot for Capstone Design
Robotics Innovatory Lab, Sungkyunkwan University
Advisor: Prof. Hyouk Ryeol Choi

Aug 2022 — Dec 2022

Built an indoor autonomous delivery robot system using ROS, Python, and a depth camera for perception and navigation.

Designed and integrated a console-controller-based teleoperation interface, connecting manual input signals to the robot's control outputs. Performed indoor environment mapping via Hector SLAM using manual control, and implemented the TEB Local Planner to enable local trajectory planning and navigation in structured indoor spaces.

3D LiDAR-Based Multiple Object Tracking for Autonomous Vehicle Perception

Mar 2021 — Feb 2022

AMLAB (Automotive Mechatronics Lab), Sungkyunkwan University

Advisor: Prof. Sung Ho Hwang

Developed a real-time 3D multiple object tracking system for autonomous vehicle perception during a research co-op at AMLAB. Designed and implemented the full LiDAR perception pipeline, including point-cloud preprocessing, RANSAC-based ground removal, voxel downsampling, ROI filtering, DBSCAN clustering, and 3D bounding box extraction. Built a Kalman-filter-based tracker with Hungarian algorithm-based data association and track lifecycle management, and integrated radar, camera, IMU, and GPS data in ROS 2 for multi-sensor perception. The system was implemented in Python using ROS 2, Open3D, NumPy, and SciPy.

Autonomous Delivery Robot for Hospital Ward Assistance

Mar 2021 — Feb 2022

AMLAB (Automotive Mechatronics Lab), Sungkyunkwan University

Advisor: Prof. Sung Ho Hwang

Developed an autonomous mobile robot system for hospital ward delivery scenarios using LiDAR-based perception, Ackermann-steering path planning, and real-time collision avoidance. Integrated sensor processing and low-level vehicle control on an NVIDIA Jetson Nano platform using Python, ROS, NumPy, and OpenCV. This project was conducted through the Sungkyunkwan University Engineering Talent Development Program and was recognized with the Excellence Award.

Development of a Simulation Drone Competition Utilizing Realistic Environments and Drone Dynamics Feb 2021 — Nov 2021

RISE LAB (Robotics and Intelligent System Engineering Lab), Sungkyunkwan University

Advisor: Prof. Hyungpil Moon

Contributed to an ICT Research Center project focused on realistic drone simulation for autonomous flight research and competition. Worked on trajectory generation and optimization based on the minimum-snap algorithm and implemented drone vibration effects induced by motor-output errors to better reflect real flight dynamics in simulation. Developed the system using Python and Unreal Engine. Received a Minister's Award.

ACTIVITIES

Sungkyunkwan University Hybrid Electric Vehicle Engineers (HEVEN)

Mar 2020 — Feb 2023

Project Leader in 2021, Perception Team Leader in 2022

- Led software development for the University Student Autonomous Driving Contest, coordinating system integration across perception and vehicle control modules.
- Received the Gold Award (1st Place) in the Technical Idea Category at the University Student Self-Made Car Contest.
- Organized and participated in the Sungkyunkwan University Autonomous Driving Hackathon.
- Led technical study sessions on autonomous driving, covering topics such as perception, planning, and control.

AWARDS

Smart Car Design and Racing Competition - Dean's Award (1st Place)

Nov 2022

Issued by Sungkyunkwan University, College of Information and Communication Engineering

- Team leader
- Designed an autonomous vehicle and implemented its driving algorithms.

Smart Car Design and Racing Competition - Excellence Award (Design)

Jun 2022

Issued by Sungkyunkwan University WE-UP

- Team leader
- Designed an autonomous vehicle and implemented its driving algorithms.

University Student Self-made Car Contest (Technical Idea Category) - Gold Award

Dec 2021

Issued by The Korean Society of Automotive Engineers (KSAE)

- Improved visibility in A-pillar blind spots using a depth camera.
- Implemented an eye-coordinate detection algorithm using MediaPipe facial landmarks.
- Processed camera images based on the detected eye position to present the driver with a viewpoint-adjusted image.

2021 ICT Colloquium - Minister's Award

Nov 2021

Issued by Ministry of Science and ICT

- Development of a simulation drone competition utilizing realistic environments and drone dynamics

Engineering Talent Development Program - Excellence Award

Nov 2021

Issued by Sungkyunkwan University

- Developed a mobile service robot for nursing hospitals.

PATENTS

Method and apparatus for evaluating object tracking performance

- Patent Application No. (KR)1020260011117, Hyundai Motor Company, Jan 20, 2026.
- *Inventors: Joohyun Lee*
- A comprehensive object tracking performance evaluation system for autonomous vehicles that receives tracking data and ground truth annotations, computes multiple evaluation metrics including association score, detection score, and classification score based on correspondence relationships between tracked and ground truth objects, applies differentiated relative weights to these metrics for weighted aggregation, and ultimately derives a composite tracking performance index to enable quantitative assessment of multi-object tracking system performance in complex driving environments.

System for processing data required for object detection based on artificial intelligence and multiple autonomous driving sensors and method implementing the same

- Patent Application No. (US)19/415,494, (KR)1020250090242, (CN)202511840140.4, Hyundai Motor Company, Dec 10, 2025.
- *Inventors: Joohyun Lee et al.*
- A hierarchical multi-level coordinate transformation system for autonomous vehicle perception that enables scalable, vehicle-agnostic integration of heterogeneous driving sensors (LiDAR, radar, and cameras) through unified sensor-to-mounting-to-vehicle coordinate conversion with comprehensive preprocessing, postprocessing, and confidence computation.

System of adaptive calibration for autonomous driving and method implementing the same

- Patent Application No. (US)19/409,994, (KR)1020250042965, (CN)202511849891.2, Hyundai Motor Company, Dec 5, 2025.
- *Inventors: Hyeonjun Lim, Joohyun Lee, Junghyoung Kwon, Minjae Kim, Chansoo Kim*
- A system and method for adaptive calibration in autonomous vehicles that uses motion and confidence scores to determine whether to perform online coordinate adjustments.

SKILLS

- **Programming:** Python, C++
- **Software:** PyTorch, OpenCV, CUDA, Docker, Git, Linux, NumPy, SciPy, ROS

LICENSES & CERTIFICATIONS

HDAT (Hyundai Motor Group Data Analytics Test) - Lv.4 (Highest)

Aug 2023

Issued by Hyundai NGV